

17 Equal volumes of aqueous KI and $0.200 \text{ mol dm}^{-3}$ of $\text{Pb}(\text{NO}_3)_2$ are mixed together to precipitate PbI_2 . Given that the K_{sp} value of PbI_2 is $8.70 \times 10^{-9} \text{ mol}^3 \text{ dm}^{-9}$, which one of the following could have been the initial concentration of KI?

A $8.70 \times 10^{-8} \text{ mol dm}^{-3}$

B $2.95 \times 10^{-4} \text{ mol dm}^{-3}$

C $5.50 \times 10^{-4} \text{ mol dm}^{-3}$

D $1.50 \times 10^{-2} \text{ mol dm}^{-3}$